

2-to-1 Binomials on Finite Fields of Odd Characteristic

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Abstract

In a recent paper [KK24], it was shown that the classification of 2-to-1 binomials over finite fields of even order is equivalent to the classification of o-monomials. In this paper, we continue the investigation of 2-to-1 binomials and study the case of odd characteristic. In particular, we show that one type of such binomials is closely related to ovals in $\text{PG}(2, q)$, which leads to their complete classification.

Key Words: binomials, preimage distribution, 2-to-1, oval in $\text{PG}(2, q)$.

The study of binomials is a classical topic in the theory of finite fields. There is a rich literature on permutation binomials, which define bijective maps on finite fields (see, for example, [AW06, MZ09, OBM22]). In this paper, we continue the investigation of 2-to-1 binomials initiated in [KK24]. Section 1 presents several general observations on maps defined by binomials, with particular emphasis on methods for describing their preimage distributions. These results are then applied in the subsequent section to the study of 2-to-1 binomials. In contrast to the case of even characteristic, there are three distinct types of 2-to-1 binomials (and, more generally, 2-to-1 maps) over finite fields of odd characteristic, as defined in Definition 2.1. We provide examples of binomials of each type. Notably, we obtain a complete classification of 2-to-1 binomials of type 2 by exploiting their strong geometric connections.

1 General properties of binomial maps

Let q be an arbitrary prime power and $d \mid q - 1$. We define

$$U_d = \{x \in \mathbb{F}_q : x^d = 1\}$$

to be the subgroup of $\mathbb{F}_q^*$ of order d , and

$$\mu_d = \{x^d : x \in \mathbb{F}_q^*\}$$

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to be the subgroup of d th powers of $\mathbb{F}_q^*$. Note that μ_d is the subgroup of order $\frac{q-1}{d}$ of $\mathbb{F}_q^*$, that is $\mu_d = U_{\frac{q-1}{d}}$. It is worth also to observe that μ_d is the image of the multiplicative homomorphism $x \mapsto x^d$ of $\mathbb{F}_q^*$, while U_d is its kernel. In general, for $k \in \mathbb{N}$ we set

$$U_k = \{x \in \mathbb{F}_q : x^k = 1\} = U_{\gcd(k, q-1)} \quad \text{and} \quad \mu_k = \{x^k : x \in \mathbb{F}_q\} = \mu_{\gcd(k, q-1)}.$$

In this paper we study maps of $\mathbb{F}_q$ defined by binomials

$$b(x) = x^s - ux^t, \tag{1}$$

where $1 \leq t < s \leq q-1$ and $u \in \mathbb{F}_q^*$. If not stated otherwise, $b(x)$ denotes the binomial given in Eq. (1).

We begin by studying elements $x, y \in \mathbb{F}_q$ such that $b(x) = b(y)$. The next observation is straightforward.

Claim 1.1. *Let $d' = \gcd(s, t, q-1)$. Then $b(x) = b(\alpha x)$ for any $x \in \mathbb{F}_q$ and any $\alpha \in U_{d'}$.*

The following result describes the preimages of 0.

Claim 1.2.

(i) *If $u \in \mu_{s-t}$, i.e. $u = u_0^{s-t}$ for some $u_0 \in \mathbb{F}_q^*$, then $b(y) = 0$ if and only if $y \in \{0\} \cup u_0 \cdot U_{s-t}$.*

(ii) *If $u \notin \mu_{s-t}$, then $b(y) = 0$ if and only if $y = 0$.*

In particular, the number of preimages of 0 under $b(x)$ is given by

$$|\{x \in \mathbb{F}_q : b(x) = 0\}| = \begin{cases} 1 + \gcd(s-t, q-1) & \text{if } u \in \mu_{s-t}, \\ 1 & \text{otherwise.} \end{cases}$$

Proof. If $0 = b(y) = y(y^{s-1} - uy^{t-1})$, then either $y = 0$ or $y^{s-t} = u$. Further, there is some $y \in \mathbb{F}_q^*$ satisfying $y^{s-t} = u$ if and only if $u \in \mu_{s-t}$, i.e., $u = u_0^{s-t}$ for some $u_0 \in \mathbb{F}_q^*$. In this case the set of solutions to $y^{s-t} = u$ is $u_0 \cdot U_{s-t}$. $\square$

Next we consider the preimages of non-zero elements. The following result on the quotient $\alpha = xy^{-1}$ of elements x, y with $b(x) = b(y)$ yields an important tool for studying binomials. Since of Claim 1.1, we consider only $\alpha \in \mathbb{F}_q^* \setminus U_{d'}$, where $d' = \gcd(s, t, q-1)$.

Lemma 1.3. *Let $d' = \gcd(s, t, q-1)$ and $y \in \mathbb{F}_q^*$, $\alpha \in \mathbb{F}_q^* \setminus U_{d'}$. Then*

$$b(\alpha y) = b(y) \tag{2}$$

if and only if

$$u^{-1} \cdot y^{s-t} = \frac{\alpha^t - 1}{\alpha^s - 1}.$$

Furthermore, if Eq. (2) holds then $\alpha \notin (U_t \cup U_s) \setminus U_{d'}$.

Proof. The assumption $b(\alpha y) = b(y)$ reduces to

$$y^s(\alpha^s - 1) = uy^t(\alpha^t - 1). \quad (3)$$

Since $y \neq 0$ and $\alpha \notin U_{d'}$, the above equation leads to the stated equivalence. Note that Eq. (3) is not satisfied when either $\alpha \in U_s$ or $\alpha \in U_t$. Indeed, in these cases exactly one of the elements $1 - \alpha^s$ or $1 - \alpha^t$ is zero since of the assumption $\alpha \notin U_{d'}$. $\square$

Motivated by Lemma 1.3, for a fixed $y \in \mathbb{F}_q^*$ we define the set

$$\begin{aligned} \mathcal{A}(y) &:= \{\alpha \in \mathbb{F}_q^* \setminus (U_s \cup U_t) : b(y) = b(\alpha y)\} \\ &= \left\{ \alpha \in \mathbb{F}_q^* \setminus (U_s \cup U_t) : u^{-1}y^{s-t} = \frac{\alpha^t - 1}{\alpha^s - 1} \right\}. \end{aligned} \quad (4)$$

Claim 1.4. *For $x, y \in \mathbb{F}_q^*$, it holds either $\mathcal{A}(x) = \mathcal{A}(y) \neq \emptyset$ or $\mathcal{A}(x) \cap \mathcal{A}(y) = \emptyset$. Furthermore, $\mathcal{A}(x) = \mathcal{A}(y) \neq \emptyset$ if and only if $y^{s-t} = x^{s-t}$, or, equivalently, x, y are in the same coset of U_{s-t} .*

Proof. By the definition if $\mathcal{A}(x) \neq \emptyset$, then $\mathcal{A}(x) = \mathcal{A}(y)$ if and only if $x^{s-t} = y^{s-t}$, or equivalently when $xy^{-1} \in U_{s-t}$. Again, the definition directly shows that $\mathcal{A}(x) \cap \mathcal{A}(y) = \emptyset$ if $y^{s-t} \neq x^{s-t}$. $\square$

Remark 1.5. For any $y \in \mathbb{F}_q^*$, the set $\mathcal{A}(y)$ is the preimage set of $u^{-1}y^{s-t}$ under the rational map $R : \mathbb{F}_q^* \setminus (U_s \cup U_t) \rightarrow \mathbb{F}_q^*$ defined by

$$R(\alpha) = \frac{\alpha^t - 1}{\alpha^s - 1}.$$

In particular,

$$|\mathcal{A}(y)| \leq s - \gcd(s, t), \quad (5)$$

since $R(\alpha) = uy^{s-t}$ has at most $s - \gcd(s, t) > t - \gcd(s, t)$ solutions α . We use here that both polynomials $\alpha^s - 1$ and $\alpha^t - 1$ are divisible by $\alpha^{\gcd(s, t)} - 1$.

Combining the previous observations we get the following central result.

Theorem 1.6. *Let $b(x) = x^s - ux^t$ with $u \in \mathbb{F}_q^*$ and $1 \leq t < s \leq q - 1$. Further, let $y \in \mathbb{F}_q^*$ be such that $z := b(y)$ is non-zero. Then $b(\alpha y) = b(y)$ if and only if $\alpha \in \mathcal{A}(y) \cup U_{d'}$, where $d' = \gcd(s, t, q - 1)$. In particular,*

$$|b^{-1}(\{z\})| = |\mathcal{A}(y)| + d'.$$

Proof. This follows directly from the definition of $\mathcal{A}(y)$ in Eq. (4) together with Claim 1.1 and Lemma 1.3. $\square$

Remark 1.7. Theorem 1.6 and Claim 1.4 show that if $z_1 = b(y_1)$ and $z_2 = b(y_2)$, where y_1 and y_2 belong to the same coset of U_{s-t} , then the number of preimages of z_1 and z_2 are equal.

In the following corollary we specify the statement of Theorem 1.6 for the permutation binomials, which are widely studied.

Corollary 1.8. *Let $b(x) = x^s - ux^t$ with $u \in \mathbb{F}_q^*$ and $1 \leq t < s \leq q - 1$. Then $b(x)$ defines a permutation on $\mathbb{F}_q$ if and only if the following three conditions are satisfied:*

- (i) $u \notin \mu_{s-t}$,
- (ii) $d' = \gcd(s, t, q - 1) = 1$,
- (iii) $\mathcal{A}(y) = \emptyset$ for every $y \in \mathbb{F}_q^*$.

Proof. The equation $b(x) = 0$ for a non-zero x is equivalent to $x^{s-t} = u$. Therefore, (i) is equivalent with $b^{-1}(\{0\}) = \{0\}$. The conditions (ii) and (iii) follow from Theorem 1.6, since $d' \geq 1$ and $1 = |\mathcal{A}(y)| + d'$ must hold when $b(x)$ is a permutation. $\square$

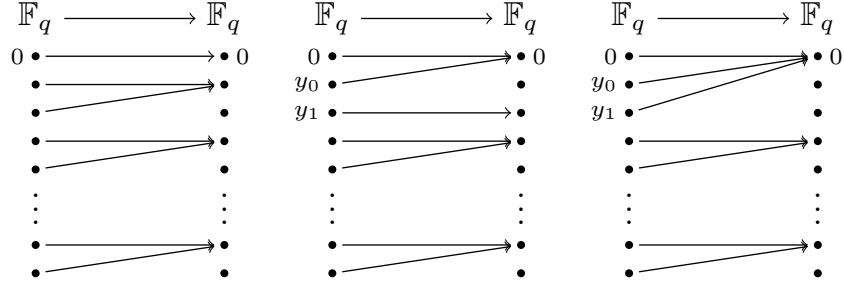
Remark 1.9. The condition $\mathcal{A}(y) = \emptyset$ for every $y \in \mathbb{F}_q^*$ is satisfied if and only if the coset $u^{-1}\mu_{s-t}$ and the image set of the rational map $R(\alpha) = \frac{\alpha^t-1}{\alpha^s-1}$ on $\mathbb{F}_q^* \setminus (U_s \cup U_t)$ are disjoint. In particular, these sets cannot be too large, forcing $\gcd(q-1, s-t)$ to be large, since $|\mu_{s-t}| = (q-1)/\gcd(q-1, s-t)$.

2 2-to-1 binomials for q odd

If q is even, then a map on $\mathbb{F}_q$ is called *2-to-1* if every element in its image set has exactly two preimages. Such binomials are characterized in [KK24]. When q is odd, a 2-to-1 map is usually defined to have a unique element in its image set with exactly one preimage, whereas all remaining elements in the image set have exactly two preimages, see for example [KP08, ZDZ⁺26]. These correspond to types 1 and 2 in Definition 2.1. We extend this notion by also calling a map 2-to-1 if 0 has exactly three preimages and all remaining elements of the image set have exactly two preimages, since in this case the map is 2-to-1 on the set of nonzero elements of $\mathbb{F}_q$.

Definition 2.1. Let q be odd and $f : \mathbb{F}_q \rightarrow \mathbb{F}_q$ with $f(0) = 0$. Then we say that

- the map f is *2-to-1 of type 1* if each non-zero element in its image set has exactly two non-zero preimages and $f(x) = 0$ only for $x = 0$.
- the map f is *2-to-1 of type 2* if there is a unique $z \neq 0$ in its image set with exactly one preimage and the remaining elements from the image set of f have exactly two preimages.
- the map f is *2-to-1 of type 3* if each non-zero element in its image set has exactly two preimages and 0 has three preimages.



(a) 2-to-1 map of type 1 (b) 2-to-1 map of type 2 (c) 2-to-1 map of type 3

By Definition 2.1, the map f is said to be *2-to-1 of type i* if 0 has exactly i preimages and f is as close as possible to a 2-to-1 map on the remaining elements of $\mathbb{F}_q^*$. The image size of a 2-to-1 function of type 1 or 2 is $(q+1)/2$, whereas for type 3 it is $(q-1)/2$. We call *exceptional* the unique element in the image of a 2-to-1 map that has either exactly one preimage or three preimages. For maps of type 1 or 3, the exceptional element is 0; for type 2 maps, the exceptional element is non-zero. By Theorem 1.6 we get:

Corollary 2.2. *Let q be odd and $b(x) = x^s - ux^t$ with $u \in \mathbb{F}_q^*$ and $1 \leq t < s \leq q-1$.*

- (i) *The binomial $b(x)$ is 2-to-1 of type 1 if and only if $\gcd(s-t, q-1) = d \geq 2$, $u \notin \mu_d$ and $|\mathcal{A}(y)| + \gcd(s, t, q-1) = 2$ for all $y \in \mathbb{F}_q^*$.*
- (ii) *The binomial $b(x)$ is 2-to-1 of type 2 with $b(y_0) = b(0) = 0$ and $b(y_1)$ exceptional (for some non-zero y_0, y_1) if and only if $\gcd(s-t, q-1) = 1$ and $|\mathcal{A}(y)| = 1$ for all $y \in \mathbb{F}_q \setminus \{y_0, y_1\}$. Moreover, in this case $u \neq 0$ can be taken arbitrary. In particular, $b(x)$ is 2-to-1 of type 2 if and only if $x^s + x^t$ is so.*
- (iii) *The binomial $b(x)$ is 2-to-1 of type 3 if and only if $\gcd(s-t, q-1) = 2$, $u \in \mu_2$, and for every $y \in \mathbb{F}_q^*$*

$$|\mathcal{A}(y)| = \begin{cases} 0 & \text{if } \gcd(s, t, q-1) = 2 \\ 1 & \text{if } \gcd(s, t, q-1) = 1. \end{cases}$$

The next proposition shows that there exist 2-to-1 binomials of each type. Note that the subgroup μ_2 consists of squares in $\mathbb{F}_q^*$ and thus $\mathbb{F}_q^* \setminus \mu_2$ is the set of non-squares.

Proposition 2.3. *Let q be odd.*

- (a) *If $b(x)$ is a permutation on $\mathbb{F}_q$, then $b(x^2)$ is a 2-to-1 of type 1 and $\mathcal{A}(y) = \emptyset$ for all $y \in \mathbb{F}_q^*$.*

- (b) Let $u \in \mathbb{F}_q^*$ such that $(u-1)(u+1) \in \mathbb{F}_q^* \setminus \mu_2$. Then the binomial $b(x) = x^{\frac{q+1}{2}} - ux$ is 2-to-1 of type 1. If $y \in \mathbb{F}_q^*$ is a square, then $\mathcal{A}(y) = \left\{ \frac{u-1}{u+1} \right\}$, otherwise $\mathcal{A}(y) = \left\{ \frac{u+1}{u-1} \right\}$.
- (c) The binomial $b(x) = x^2 - 2x$ is 2-to-1 of type 2. For $y \in \mathbb{F}_q^* \setminus \{2\}$ we have $\mathcal{A}(y) = \left\{ \frac{2-y}{y} \right\}$.
- (d) Let $-u$ be a non-square in $\mathbb{F}_q$. The binomial $b(x) = x^{q-2} - ux$ is 2-to-1 of type 1 if $q \equiv 1 \pmod{4}$ and of type 3 if $q \equiv 3 \pmod{4}$. Moreover, $\mathcal{A}(y) = \{-u^{-1}y^{q-3}\}$ for all $y \in \mathbb{F}_q^*$.
- (e) The binomial $b(x) = x^{\frac{q-1}{2}+1} + ux^{\frac{q-1}{2}-1}$ is 2-to-1 of type 3 if $q \equiv 3 \pmod{4}$ and $u \in \mu_2$. Moreover, $\mathcal{A}(y) = \emptyset$ for all $y \in \mathbb{F}_q^*$.

Proof. (a) is straightforward. (b) Suppose for fixed $y \in \mathbb{F}_q^*$ and $\alpha \in \mathbb{F}_q^* \setminus U_{(q+1)/2}$ we have $b(y) = b(\alpha y)$. By Lemma 1.3 this is equivalent to

$$u^{-1}y^{\frac{q-1}{2}} = \frac{\alpha - 1}{\alpha^{\frac{q+1}{2}} - 1}. \quad (6)$$

Note that $\alpha^{\frac{q+1}{2}} = \alpha$ if $\alpha \in \mu_2$ and $-\alpha$ if $\alpha \notin \mu_2$. In the same vein, note that $y^{\frac{q-1}{2}} = 1$ if $y \in \mu_2$ and -1 if $y \in \mathbb{F}_q^* \setminus \mu_2$. Eq. (6) implies that α cannot be a square, since that would force $u \in \{-1, 1\}$. Similarly, if $-1 \notin U_{\frac{q+1}{2}}$, then α cannot be -1 for the same reason. Lemma 1.3 prohibits $\alpha \in U_{\frac{q+1}{2}}$, so $\alpha \neq -1$. Then Eq. (6) is equivalent to

$$\alpha = \frac{1 - u^{-1}y^{\frac{q-1}{2}}}{1 + u^{-1}y^{\frac{q-1}{2}}} = \frac{u - y^{\frac{q-1}{2}}}{u + y^{\frac{q-1}{2}}}.$$

Thus every image of b except for zero has exactly two preimages.

(c) Note that $b(x) = (x-1)^2 - 1$ and hence every element $\neq -1$ from its image set has exactly 2 preimages, whereas the only preimage of -1 is 1. The second statement follows from a short calculation.

(d) For fixed $y \in \mathbb{F}_q^*$ and $\alpha \in \mathbb{F}_q^* \setminus U_{q-2}$ by Lemma 1.3 we have $b(y) = b(\alpha y)$ if and only if

$$u^{-1}y^{q-3} = \frac{\alpha - 1}{\alpha^{q-2} - 1} = -\alpha,$$

so $\alpha = -\frac{1}{uy^2}$. Since $-u$ is not a square we have $\alpha \neq 1$, so each image of $b(x)$ in $\mathbb{F}_q^*$ has exactly 2 preimages. The type of $b(x)$ follows from Claim 1.2 by the fact that $-1 \in \mu_2$ if $q \equiv 3 \pmod{4}$ and $-1 \notin \mu_2$ if $q \equiv 1 \pmod{4}$.

(e) The exponents of the binomial satisfy $d' = \gcd(\frac{q+1}{2}, \frac{q-3}{2}, q-1) = 2$. By Lemma 1.3 for fixed $y \in \mathbb{F}_q^*$ and $\alpha \in \mathbb{F}_q^* \setminus U_2$ we have $b(y) = b(\alpha y)$ if and only if

$$u^{-1}y^2 = \frac{\alpha^{\frac{q-1}{2}} \frac{1}{\alpha} - 1}{\alpha^{\frac{q-1}{2}} \alpha - 1}.$$

If $\alpha^{\frac{q-1}{2}} = 1$, this equation becomes $u^{-1}y^2 = -\alpha^{-1}$. There are no solutions in this case since $u^{-1}y^2$ is a square and $-\alpha^{-1}$ is a non-square (note that $-1 \notin \mu_2$). Similarly, there are no solutions for $u^{-1}y^2 = \alpha^{-1}$ in the case $\alpha^{\frac{q-1}{2}} = -1$. Thus, $\mathcal{A}(y) = \emptyset$ for all $y \in \mathbb{F}_q^*$ and $b(x)$ is 2-to-1 by Theorem 1.6. The type follows from Corollary 2.2. $\square$

Type 1 and type 2 binomials appear quite similar, since in both cases the exceptional element has a unique preimage. (However, Section 3 shows that type 2 binomials are highly special due to their geometric interpretation.) The next result shows that there are no 2-to-1 binomials for which the exceptional element is non-zero and has (at least) three preimages.

Proposition 2.4. *Let q be odd, let $1 \leq t < s \leq q-1$ and let $b(x) = x^s - ux^t$. Suppose there exists a unique non-zero element $z \in \text{Image}(b)$ such that every element $z' \in \text{Image}(b) \setminus \{z\}$ has exactly two preimages under b . Then z has exactly one preimage. In particular, $b(x)$ is a 2-to-1 binomial of type 2.*

Proof. By Claim 1.2, we have $\gcd(s-t, q-1) = 1$ and hence $\gcd(s, t, q-1) = 1$ too. Denote by $y_0 \in \mathbb{F}_q^*$ the unique non-zero element satisfying $b(y_0) = b(0) = 0$. Further, let Ξ be the set of preimages of z . Since q is odd, the size of Ξ must be odd too. Suppose Ξ has at least 3 elements. Consider the set

$$\mathcal{A} := \bigcup_{y \in \mathbb{F}_q^* \setminus \{y_0\}} \mathcal{A}(y).$$

Note that we have $|\mathcal{A}(y)| = 1$ for every non-zero $y \neq y_0$ not belonging to Ξ , whereas $|\mathcal{A}(y)| \geq 2$ for every $y \in \Xi$. Moreover, by Claim 1.4 we have $\mathcal{A}(x) \cap \mathcal{A}(y) = \emptyset$ for distinct $x, y \in \mathbb{F}_q^*$. Hence we get

$$|\mathcal{A}| = \sum_{y \in (\mathbb{F}_q^* \setminus \{y_0\}) \setminus \Xi} |\mathcal{A}(y)| + \sum_{y \in \Xi} |\mathcal{A}(y)| \geq q-2-3 + 3 \cdot 2 = q+1,$$

a contradiction to $\mathcal{A} \subseteq \mathbb{F}_q^*$. $\square$

3 The classification of 2-to-1 binomials of type 2

In this section, we study 2-to-1 binomials of type 2. These binomials can be regarded as the natural counterparts of 2-to-1 binomials in even characteristic, both with respect to their algebraic properties and their characterization through finite geometry. This geometric viewpoint enables a complete classification of 2-to-1 binomials of type 2. The results presented here are motivated by [KK24].

Fix integers t, s with $1 \leq t < s \leq q - 1$. By Corollary 2.2(b), if there exists a nonzero element $u \in \mathbb{F}_q$ such that the binomial $b(x) = x^s - ux^t$ is 2-to-1 of type 2 on $\mathbb{F}_q$, then for every nonzero $u' \in \mathbb{F}_q$ the binomial $x^s - u'x^t$ is also 2-to-1 of type 2. This shows that in such a case it is enough to consider the binomial $x^s + x^t$. By Corollary 2.2 such exponents s, t necessarily satisfy $\gcd(s-t, q-1) = 1$. The following lemma establishes an additional necessary condition on the exponent pair (s, t) .

Lemma 3.1. *Let q be odd and $1 \leq t < s \leq q - 1$. A 2-to-1 binomial $b(x) = x^s - ux^t$ of type 2 on $\mathbb{F}_q$ satisfies either $\gcd(s, q-1) = 2$ and $\gcd(t, q-1) = 1$ or $\gcd(s, q-1) = 1$ and $\gcd(t, q-1) = 2$.*

Proof. Denote by $y_0 \in \mathbb{F}_q^*$ the unique non-zero element satisfying $b(y_0) = b(0) = 0$ and set y_1 to be the preimage of the exceptional element of b . For any element $y \in \mathbb{F}_q \setminus \{0, y_0, y_1\}$ exists a unique $\alpha \in \mathbb{F}_q^* \setminus (U_s \cup U_t)$ such that $b(y) = b(\alpha y)$, equivalently $uy^{s-t} = R(\alpha)$ using Lemma 1.3 and Remark 1.5. Note that $y \mapsto uy^{s-t}$ defines an injective map for $y \in \mathbb{F}_q \setminus \{0, y_0, y_1\}$, and thus the rational map $R : \mathbb{F}_q^* \setminus (U_s \cup U_t) \rightarrow \mathbb{F}_q^*$ must have at least $|\mathbb{F}_q \setminus \{0, y_0, y_1\}| = q - 3$ elements in its image set. Since obviously

$$|\{R(\alpha) : \alpha \in \mathbb{F}_q^* \setminus (U_s \cup U_t)\}| \leq |\{\alpha \in \mathbb{F}_q^* \setminus (U_s \cup U_t)\}|$$

and the latter set has size $q - \gcd(s, q-1) - \gcd(t, q-1)$, it must hold

$$\gcd(s, q-1) + \gcd(t, q-1) \leq 3,$$

completing the proof. $\square$

Next we show that 2-to-1 binomials of type 2 are closely related to ovals in finite Desarguesian projective planes of odd order. This connection yields a complete classification of 2-to-1 binomials of type 2. The result complements the case of even characteristic, where the classification of 2-to-1 binomials is equivalent to that of o-monomials, that is, hyperovals induced by monomials [KK24].

The finite Desarguesian projective plane of order q is the set of subspaces of $\mathbb{F}_q^3$, abbreviated by $\text{PG}(2, q)$. Its points are the one-dimensional subspaces of $\mathbb{F}_q^3$, which we denote by

$$(x_1, x_2, x_3) := \left\langle \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \right\rangle.$$

The lines of the plane are the two-dimensional subspaces of $\mathbb{F}_q^3$, which can be represented as orthogonal complements of points. Incidence of points and lines is defined via the inclusion in $\mathbb{F}_q^3$. There are $q^2 + q + 1$ points and accordingly also $q^2 + q + 1$ lines in $\text{PG}(2, q)$ in total.

An oval of $\text{PG}(2, q)$ is a set of $q + 1$ points of $\text{PG}(2, q)$, of which no three are collinear, e.g. no three points of the oval are on a common line. A

standard reference for projective planes and ovals is [Hir98]. Every point of an oval has a unique tangent line meeting the oval only in that point. This is reminiscent of the fact that the exceptional elements of 2-to-1 binomials of type 2 have exactly one preimage.

Simple examples of ovals are conics, the zero set of a non-singular homogeneous quadratic equation over $\mathbb{F}_q^3$. For odd characteristic Segre showed the remarkable converse statement [Seg55].

Theorem 3.2 (Segre's Theorem). *If q is odd, then any oval of $\text{PG}(2, q)$ is a conic.*

Thus, in odd characteristic, ovals are well understood, which leads to a complete classification of 2-to-1 binomials of type 2, as we show in the remainder of this section.

Next lemma shows that exponents s, t of a 2-to-1 binomial define an oval in $\text{PG}(2, q)$. By Lemma 3.1 either $\gcd(s, q-1) = 1$ or $\gcd(t, q-1) = 1$. Without loss of generality we assume $\gcd(t, q-1) = 1$.

Lemma 3.3. *Let q be odd and $b(x) = x^s + x^t$ be 2-to-1 of type 2 with $\gcd(t, q-1) = 1$. Then the set*

$$\mathcal{O}(s, t) := \{(1, y^t, y^s) : y \in \mathbb{F}_q\} \cup \{(0, 0, 1)\} \subset \text{PG}(2, q)$$

is an oval.

Proof. Notice first that $|\mathcal{O}(s, t)| = q + 1$. Indeed, if $v^s = w^s$ and $v^t = w^t$ for a pair $(v, w) \in (\mathbb{F}_q^*)^2$ with $v \neq w$, then for the binomial $b_1(x) := x^s - x^t$ we get

$$b_1\left(\frac{v}{w}\right) = b_1(1) = b_1(0) = 0,$$

a contradiction, since the binomial $b_1(x)$ is 2-to-1 of type 2 as well.

Denote by $l_{a,b} := \left\langle \begin{pmatrix} a \\ b \\ 1 \end{pmatrix} \right\rangle^\perp$, $l_a := \left\langle \begin{pmatrix} a \\ 1 \\ 0 \end{pmatrix} \right\rangle^\perp$, and $l_\infty := \left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle^\perp$ for $a, b \in \mathbb{F}_q$ the lines of $\text{PG}(2, q)$. The point $(0, 0, 1)$ is not on $l_{a,b}$ and $(1, y^t, y^s) \in \mathcal{O}(s, t)$ with $y \in \mathbb{F}_q$ is on $l_{a,b}$ if and only if

$$a + by^t + y^s = 0. \tag{7}$$

Eq. (7) has at most 2 solutions y for every $b \neq 0$, since the binomial $y^s + by^t$ is 2-to-1 of type 2. If $b = 0$, then Eq. (7) has either two or zero solutions since $\gcd(s, q-1) = 2$ by Lemma 3.1.

On the lines l_a with $a \in \mathbb{F}_q$ we find the point $(0, 0, 1)$ and the point $(1, y^t, y^s)$, with y being the unique solution of

$$a + y^t = 0.$$

Finally, l_∞ contains only $(0, 0, 1)$ from $\mathcal{O}(s, t)$. Thus $\mathcal{O}(s, t)$ is an oval. $\square$

Remark 3.4. Lemmas 3.1 and 3.3 admit alternative proofs based on a combinatorial line-counting argument, see [Oer24, Theorem 4.2.8]. It is also worth to note that the converse of Lemma 3.3 is true as well: $\mathcal{O}(s, t)$ is an oval if and only if $b(x) = x^s + x^t$ is 2-to-1 of type 2. The complete prove in the full version of the paper.

Theorem 3.5. *Let q be odd and $b(x) = x^s - ux^t$ be a 2-to-1 binomial of type 2 with $\gcd(t, q-1) = 1$. Then $\frac{s}{t} \equiv 2 \pmod{q-1}$. In particular, $b(x)$ can be traced back to $x^2 - ux$ by the substitution $x \mapsto x^{\frac{1}{t}}$. (Here $1/t$ denotes the inverse of t modulo $q-1$.)*

Proof. Using the substitution $x \mapsto x^{\frac{1}{t}}$, we have

$$\mathcal{O}(s, t) = \mathcal{O}\left(\frac{s}{t}, 1\right) = \left\{\left(1, y, y^{\frac{s}{t}}\right) : y \in \mathbb{F}_q\right\} \cup \{(0, 0, 1)\} =: \mathcal{O},$$

By Segre's Theorem $\mathcal{O}$ is a conic, so let

$$G(x, y, z) := ax^2 + by^2 + cz^2 + fyz + gzx + hxy$$

with $a, b, c, f, g, h \in \mathbb{F}_q$ be the associated quadratic form.

Since $(0, 0, 1) \in \mathcal{O}$, we have $c = 0$ and from $(1, 0, 0) \in \mathcal{O}$ it follows that $a = 0$. Letting k be the least positive residue of $\frac{s}{t} \pmod{q-1}$, define

$$F(x) := G(1, x, x^k) = bx^2 + fx^{k+1} + gx^k + hx \in \mathbb{F}_q[x].$$

Note that $\gcd(k, q-1) = 2$ follows from $\gcd(s, q-1) = 2$.

Then $F(t) = 0$ for all $t \in \mathbb{F}_q$, so either $\deg F = q$ or F is the zero polynomial. In the first case, we must have $k = q-1$. Unless $q = 3$, this would imply $\gcd(k, q-1) = q-1 \neq 2$. If $q = 3$, we have $k = 2$ and are done.

Thus only the case $F(x) = 0$ remains. As G is non-singular, not all coefficients vanish. This rules out $k \geq 3$, as all monomials in F would have a different degree. And finally, $\gcd(1, q-1) = 1$, so $k = 1$ cannot happen either. Therefore, $k = 2$. $\square$

We summarize our results on the 2-to-1 binomials of type 2 in the following theorem, which gives the complete classification of such binomials.

Theorem 3.6. *Let q be odd and let $s, t \in \mathbb{N}$. The binomial*

$$b(x) = x^s - ux^t$$

is 2-to-1 of type 2 on $\mathbb{F}_q$ if and only if $u \in \mathbb{F}_q^$ is arbitrary and the pair of exponents (s, t) satisfies one of the following conditions:*

$$\begin{cases} \gcd(t, q-1) = 1 \text{ and } s \equiv 2t \pmod{q-1}, \\ \gcd(s, q-1) = 1 \text{ and } t \equiv 2s \pmod{q-1}. \end{cases}$$

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